

Situately : Embodied Conversational Agent With LLM-Powered Nonverbal Behavior and Object Interaction

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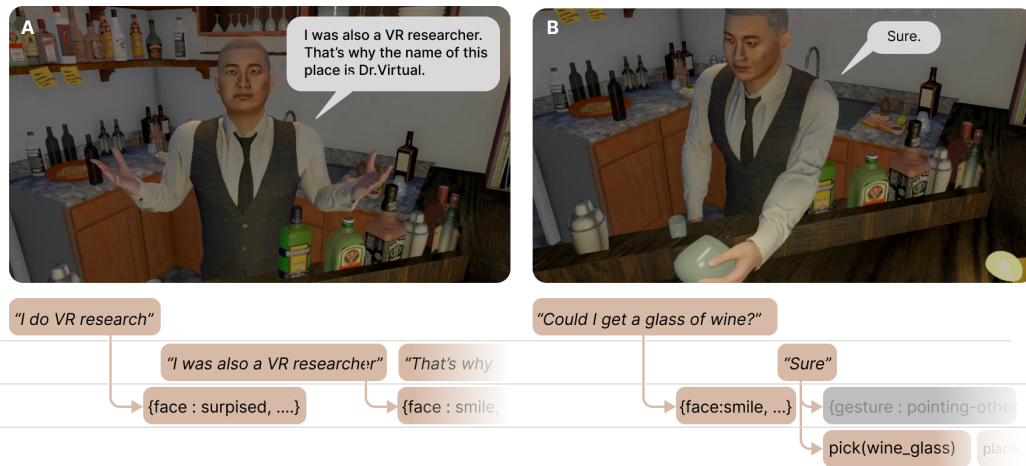


Figure 1: Examples of nonverbal behavior and object interaction aligned with verbal context. (A) For each utterance, nonverbal behaviors are planned and expressed to align with the verbal content. (B) From the verbal response, the object interaction controller is triggered as a tool, and a sequence of actions is planned and executed accordingly.

ABSTRACT

Embodied conversational agents possess virtual bodies in virtual environments, yet often lack socially and spatially plausible bodily behavior during interaction. We present a real-time, concurrent framework for embodied conversational agents that controls and coordinates nonverbal behavior and spatially grounded actions beyond verbal interaction alone. Built on large language models and ongoing conversational context, our system enables multimodal control by generating structured outputs for avatar-level nonverbal expression and object interaction. These structured outputs are directly mapped to real-time avatar animations, allowing users to perceive the agent's nonverbal expressions and object interactions in a shared space. We demonstrate our approach through an interactive virtual environment demo, showing how it supports richer embodied interaction and increases perceived animacy.

Index Terms: Embodied Conversation Agent, Embodied-AI, Human-AI Interaction.

1 INTRODUCTION

Embodied conversational agents (ECAs) are expected to go beyond language by expressing social cues and interacting with virtual objects through spatially grounded bodily behavior. In situated interaction, users often expect appropriate nonverbal responses, such as gaze or nodding while speaking or listening, and may expect the agent to interact with objects. When these expectations are unmet, even fluent verbal responses fail to sustain a sense of social

presence, revealing the limitations of language-only approaches to embodied interaction.

Meanwhile, nonverbal response generation and agent-object interaction have long been central topics in computer graphics and animation research. However, prior approaches typically rely on large, computationally intensive models or offline pipelines, which limits their applicability to real-time interaction research requiring low latency, rapid iteration, and experimental control. As a result, these methods are difficult to employ in studies that aim to examine human experience during live social interaction.

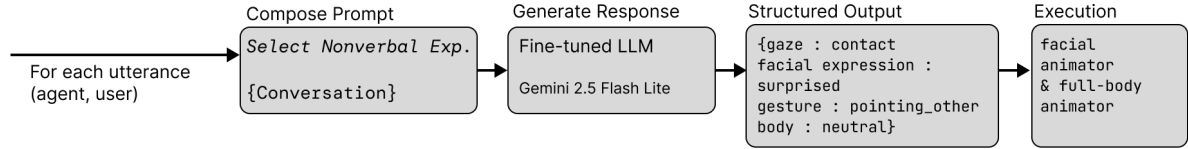
As an alternative to such heavyweight approaches, LLM-infused methods have recently gained attention [1, 2, 3]. Large language models have demonstrated the ability to predict context-appropriate actions by leveraging commonsense knowledge embedded in language while operating under real-time latency constraints. This capability enables the study of users in interactive settings that were previously difficult to address with deep neural architectures. However, prior work often treats nonverbal behavior and object interaction as independent technical components rather than as coordinated elements of social interaction.

In this demo, we address this gap by integrating nonverbal behavior and object interaction within embodied conversational agents. We leverage large language models to generate structured outputs that drive nonverbal expression and object interaction. These outputs are executed through real-time animation controllers. We refer to this system as *Situately*, emphasizing its focus on situated interaction.

We demonstrate the system through a bartender-style embodied agent situated in a virtual bar environment, where users engage in open-ended conversation and interact with the surrounding space. This scenario illustrates how integrated nonverbal behavior and object interaction support expressive and situated interaction while remaining practical and reproducible for ECA studies.

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(A) Nonverbal Behavior Controller



(B) Object Interaction Controller

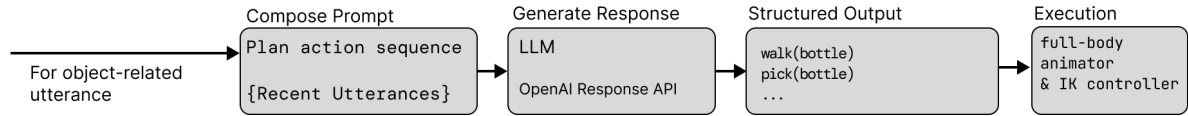


Figure 2: Overview of the nonverbal behavior and object interaction controllers. (A) For each user or agent utterance, the nonverbal behavior controller selects context-appropriate nonverbal expressions using a fine-tuned language model and outputs structured representations for avatar control. (B) For object-related utterances, the object interaction controller plans an action sequence using a language model and produces structured action-object commands for execution.

2 DESIGN & IMPLEMENTATION

Verbal Interaction The verbal interaction module is implemented using an audio-to-audio conversational service based on the OpenAI Realtime API, enabling low-latency, real-time interaction without explicit speech-to-text or text-to-speech pipelines. Lip-sync was driven solely by voice amplitude. Prompts specify the agent’s personality, situational role, conversational skills, and event triggers. Using the same system in a separate experiment, the agent’s audio-to-audio latency was 2.82 s (SD = 2.76).

Nonverbal Behavior Controller The nonverbal behavior controller operates within the conversational thread, selecting context-appropriate nonverbal behaviors based on the current interaction state (see Figure 2A). Nonverbal expressions include gaze, facial expressions, gestures, and head movements, which are executed in compositional combinations and triggered both during the agent’s speaking turns and while listening to the user. This process is implemented using a fine-tuned language model (Gemini 2.5 Flash Lite, trained via VertexAI) trained on expert-annotated 10-minute conversational segments (N = 21) to align model outputs with human-selected behaviors. Output behaviors were executed after filtering to remove clearly mismatched cases (e.g., a co-speech gesture during listening). The latency for nonverbal response was 0.75 s (SD = 0.25) in a separate experiment using the same system. In the same experiment, participants (N = 36) reported significantly higher perceived animacy with nonverbal behaviors than without ($t(252) = 2.59, p = .031$), while no significant improvement was observed in social presence.

Object Interaction Controller The object interaction controller acquires information about objects present in the virtual scene and enables context-appropriate object interaction (see Figure 2B). It is invoked via tool-calling¹ from the verbal interaction module when the agent needs to perform object-related actions during conversation. Once invoked, the controller generates an action-object plan that specifies a sequence of actions and their corresponding target objects using a language model (Gemini 2.5 Flash Lite). The available actions were predefined as walk, pick, put, and pour, and interactable objects were also predefined within the scene. The generated action plan is then executed in the application through coordinated control of an inverse kinematics controller and a full-body animation system. When conflicts arose with nonverbal behaviors, nonverbal expressions were suppressed. No external failure detection or recovery loop was implemented,

¹Tools in the OpenAI Realtime API <https://platform.openai.com/docs/guides/tools>

and context-inappropriate actions were addressed through user corrective actions.

3 DEMONSTRATION

We demonstrate our system through an interactive virtual reality scenario in which participants engage in open-ended conversation with a virtual bartender in a bar environment. The agent generates verbal responses, nonverbal expressions, and object interactions in real time, executing them as context-appropriate animations with a Avaturn avatar. The demonstration is implemented in Unity Engine for Meta Quest Pro VR headset.

4 CONCLUSION & FUTURE WORK

We presented a real-time framework for embodied conversational agents that integrates verbal interaction, nonverbal behavior, and object interaction under concurrent control. By coordinating multiple LLM-driven controllers within real-time constraints, the system enables context-appropriate agent responses that remain coherent and physically grounded. While the system increased perceived animacy, this did not translate into a broader improvement in social experience. Future work is needed to examine whether increasing behavioral diversity and animation quality can bridge this gap.

ACKNOWLEDGMENTS

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